

9 D

$D = kv^2$, where k is a constant of proportionality

At 12 m s^{-1} , the driving force provided by the 2 engines is equal and opposite to the total drag.

$$P = Dv$$

$$36000 \times 2 = k(12)^3$$

$$k = \frac{720000}{(12)^3}$$

If only 1 engine is on, the new maximum speed v' can be found by:

$$360000 = \frac{720000}{(12)^3} (v')^3$$

$$\begin{aligned} v' &= \sqrt[3]{\frac{360000}{720000} (12^3)} \\ &= \frac{12}{\sqrt[3]{2}} = 9.5 \text{ m s}^{-1} \end{aligned}$$

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